MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, 11747101

RANGE AND RANGE RATE GENERATOR

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers one type of Range and Range Rate Generator printed circuit board, herein referred to as the "Range and Range Rate Generator" which forms a part of SCG Servo Test Set 11747100.

2. APPLICABLE DOCUMENTS.

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction, M18, Manufacturing Process and Inspection of

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

F11747101	Generator, Range & Range Rate
F11747103	Schematic, Generator, Range & Range Rate
D10554938	Connector

PACKAGING DATA SHEET

11747101	Packaging of Generator, Range & Range Rate

(Copies of specifications, standards, drawings, and packaging data sheets, required by contractors in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Fabrication. The Range and Range Rate Generator shall be manufactured in accordance with Drawing F11747101 and drawings pertaining thereto.

3.2 General specification. Where applicable, the Range and Range Rate Generator shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence.
b. Dimensions and tolerances.
c. Inorganic protective surface finishings.
d. Part identification and marking.
e. Workmanship.
f. Soldering electrical connections.

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3.2.2 MIL-C-46357.

a. Etched circuit assemblies.
b. Wiring.
c. Threaded fasteners.
d. Marking.
e. Finishes.

3.3 Environmental service conditions.

3.3.1 Vibration. The Range and Range Rate Generator shall be mounted on the vibration table in normal service position. The Range and Range Rate Generator shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

TABLE I. Vibration

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. The Range and Range Rate Generator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 $\pm$ 1 milliseconds. The G value of each impulse shall be within $\pm$ 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating. The Range and Range Rate Generator shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

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3.3.3.2 Nonoperating. The Range and Range Rate Generator shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The Range and Range Rate Generator shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Nonoperating (storage). The Range and Range Rate Generator shall not suffer permanent damage when subjected to the temperature extremes of -80°F and +185°F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity-nonoperating. The Range and Range Rate Generator shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.

3.4.1 Continuity. Each conductor shall be continuous electrically as specified on the schematic diagram F11747103.

3.4.2 Resistance. Resistance shall be 1000 ohms ± 10% between the following terminals.

From	To
P1-11	P1-30
P1-45	P1-30

3.5 Performance.

3.5.1 Input signal. For the requirements of 3.5.3 through 3.5.8, logic input and output levels shall be as follows:

Logic 0: 0.0 ± 0.5 Vdc Level

Logic 1: 4.0 ± 1.2 Vdc Level

↓ : Momentary Transition from Logic 1 to Logic 0

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3.5.2 Input power. For the requirements of 3.5.3 through 3.5.8, input power shall be provided in accordance with the test setup illustrated in Figure 1.

3.5.3 Output pin 3. The output, with a 1K load between pin 3 and pin 22 shall be +12 Vdc ± 5.0 percent between pin 3(+) and pin 22(-).

3.5.4 Range output. With a Logic 0 applied to P1-11 and with P1-45, P1-5, P1-6 at Logic 0, the range output pulses shall have the periods specified in Table II. All outputs shall be positive 50 percent square waves from 0V to 4.0 ± 1.2 Vdc. P1-24 shall be used as return for all signals.

TABLE II. Range Outputs

Description	TP	Characteristic	Terminal
5M	1	16.6 ± .4 ms	P1-52
10M	2	33.2 ± .8 ms	P1-51
20M	3	66.4 ± 1.6 ms	P1-55
40M	4	132.8 ± 3.2 ms	P1-56
80M	5	265.6 ± 6.4 ms	P1-54
160M	6	531.2 ± 12.8 ms	P1-53
320M	7	1.06 ± .03 SEC	P1-49
640M	8	2.12 ± .06 SEC	P1-50
1280M	9	4.24 ± .12 SEC	P1-47
2560M	10	8.48 ± .24 SEC	P1-46

3.5.5 Range Rate output. With a Logic 0 applied to P1-45 and with P1-11, P1-5, P1-6 at Logic 0, the range rate output pulses shall have the periods specified in Table III. All outputs shall be positive 50% square wave from 0V to 4.0 ± 1.2 Vdc. P1-24 shall be used as return for all signals.

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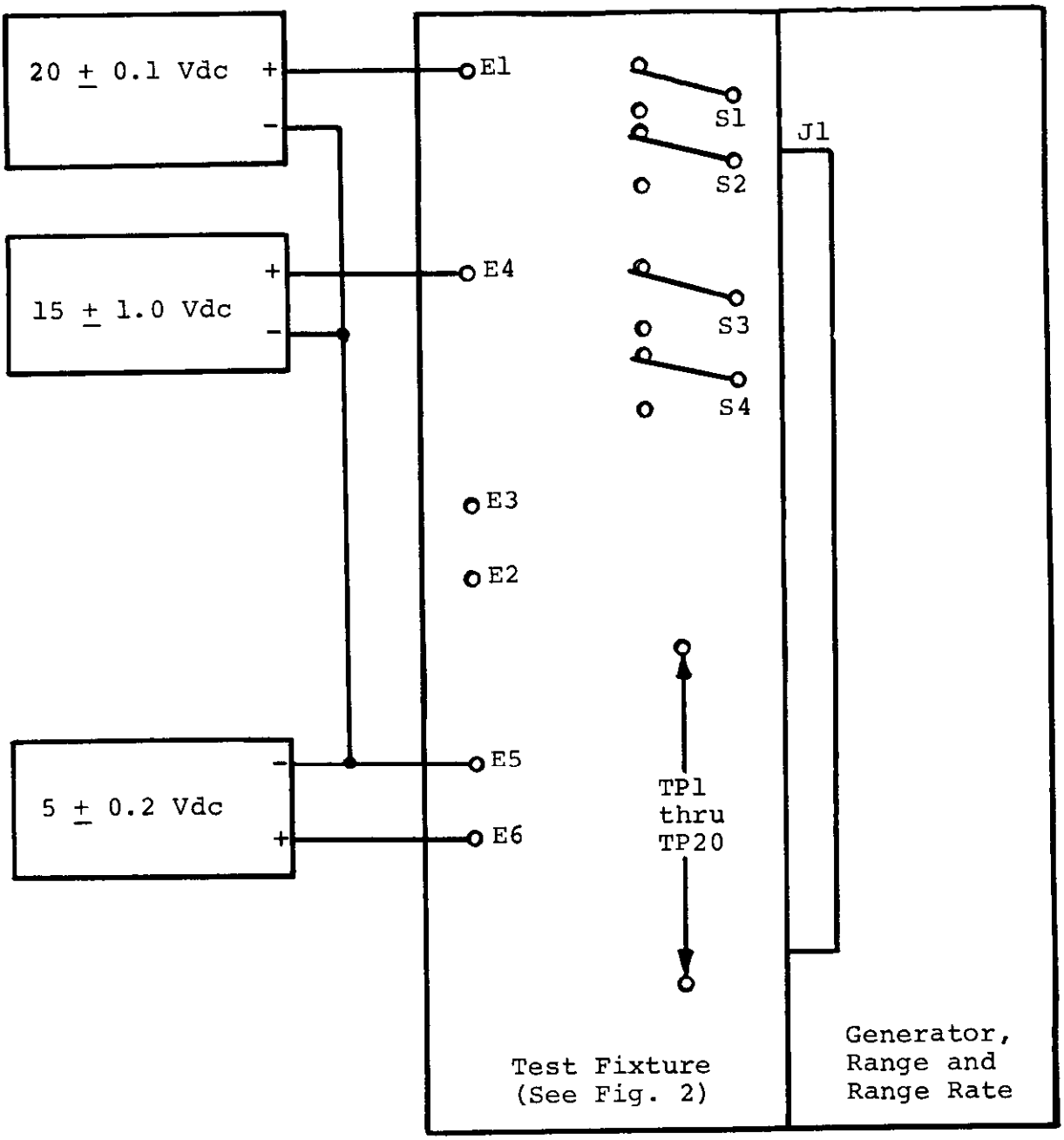


FIGURE 1. Range and Range Rate Generator Test

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TABLE III. Range Rate Outputs

Description	TP	Characteristic	Terminal
2.5 M/S	11	132.8 ± 3.2 ms	P1-19
5 M/S	12	265.6 ± 6.4 ms	P1-17
10 M/S	13	531.2 ± 12.8 ms	P1-15
20 M/S	14	1.06 ± .03 SEC	P1-41
40 M/S	15	2.12 ± .06 SEC	P1-43
80 M/S	16	4.24 ± .12 SEC	P1-40
160 M/S	17	8.48 ± .24 SEC	P1-39

3.5.6 Range count signals.

3.5.6.1 Up count ranges. With the input Logic Levels applied as specified in Table IV, Step 1, two consecutive output signals as specified in Table II shall first be out of phase for the first part of the run indicating a count up direction control and then in phase for the second part of the run indicating a count down direction control. The time for the first part of the run shall be 8.0 ± .4 seconds for the 5M and 10M outputs of Table II. The time for an entire run shall be 16.0 ± .8 seconds.

3.5.6.2 Up count range rate. With the input Logic Levels applied as specified in Table IV, Step 2, two consecutive output signals as specified in Table III shall be in phase for 8.1 ± .1 SEC indicating a count up direction control and then out of phase for 8.1 ± .1 SEC indicating a count down direction control.

3.5.6.3 Down count range. With the input Logic Levels applied as specified in Table II, Step 3, two consecutive output signals as specified in Table II shall be in phase for the first part of the run indicating a count down direction control and then out of phase for the second part of the run indicating a count up direction control. The time for the first part of the run shall be 8.0 ± .4 seconds for the 5M and 10M outputs of Table II. The time for the entire run shall be 16.0 ± .8 seconds.

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3.5.7 Range rate count signals.

3.5.7.1 Up count range rate. With the input Logic Levels applied as specified in Table IV, Step 2, two consecutive output signals as specified in Table III shall be out of phase for the first part of the run indicating a count up direction control and then in phase for the second part of the run indicating a count down direction control. The time for the first part of the run shall be 8.0 ± .4 seconds for the 2.5 M/S and 5 M/S outputs of Table III. The time for the entire run shall be 16.0 ± .8 seconds.

3.5.7.2 Down count range rate. With the input Logic Levels applied as specified in Table IV, Step 4, two consecutive output signals as specified in Table III shall be in phase for the first part of the run indicating a count down direction control and then out of phase for the second part of the run indicating a up count direction control. The time for the first part of the run shall be 8.0 ± .4 seconds for the 2.5 M/S and 5 M/S outputs of Table III. The time for the entire run shall be 16.0 ± .8 seconds.

TABLE IV. Input Logic Levels

Step	P1-5	P1-6	P1-11	P1-45
1	0	1	↓*	1
2	0	1	1	↓*
3	1	0	↓*	1
4	1	0	1	↓*

* Not to be toggled till ready to make run.

3.5.8 Sign-Sign output. In Paragraphs 3.5.7.1 and 3.5.7.2, whenever two consecutive output signals of Table III are out of phase, indicating a count up condition, P1-13 shall be a Logic 1 and P1-9 shall be a Logic 0. Whenever two consecutive signals are in phase, indicating a count down condition, P1-9 shall be a Logic 1 and P1-13 shall be a Logic 0.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.4 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three Range & Range Rate Generator Boards manufactured in the same manner, using the same materials, equipment,

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processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in the following table, as sequenced.

First Article Test Procedures

	Characteristic	Environment	Requirements	Test Procedures
1.	Vibration	4.8.1.1	3.4.1 through 3.5.8	4.8.2 through 4.8.3
2.	Shock	4.8.1.2	3.4.1 through 3.5.8	4.8.2 through 4.8.3
3.	Altitude (operating and non-operating)	4.8.1.3	3.5.3 through 3.5.8	4.8.3.1 through 4.8.3.7
4.	High Temprature	4.8.1.4.1	3.4.1.2 through 3.5.7	4.8.2.1 through 4.8.3.7
5.	Low Temperature	4.8.1.4.2	3.4.1 through 3.5.8	4.8.2.1 through 4.8.3.7
6.	Humidity	4.8.1.5	3.4 through 3.5.8	4.8.2 through 4.8.3

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement. Rejected assemblies must be replaced with new first article assemblies for repeated first article testing. Rejected assemblies shall become property of the Government and no longer used for first article, production items, or canabalization.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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4.5 Inspection provisions.

4.5.1 Submission of product. Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.5.2 Examination and tests. Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables V and VI herein. Examinations and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE V. Classification of Defects

100% Inspection

No.	Characteristic	Requirement	Test Procedure
101.	Continuity	3.4.1	4.6.2.1
102.	Resistance	3.4.2	4.6.2.2
103.	Output Pin 3	3.5.3	4.6.3.2
104.	Range Output	3.5.4	4.6.3.3
105.	Range Rate Output	3.5.5	4.6.3.4
106.	Range Count Signals	3.5.6	4.6.3.5
107.	Range Rate Count Signals	3.5.7	4.6.3.6
108.	Sign-$\overline{\text{Sign}}$ Output	3.5.8	4.6.3.7

4.5.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.6 Special sampling. One Range and Range Rate Generator out of the first 25 produced and one out of each 25 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table V and shall then meet the requirements and tests in Table VI.

TABLE VI. Special Sampling

Characteristic	Requirement	Test Procedure
Fabrication	3.1	Applicable Dwg. Visual - Tactile
General Specification	3.2	MIL-F-13926, MIL-C-46357 Visual - Tactile
Vibration	3.3.1	4.6.1.1
Shock	3.3.2	4.6.1.2
Altitude - Operating	3.3.3.1	4.6.1.3
Altitude - Nonoperating	3.3.3.2	4.6.1.3
Temperature	3.3.4	4.6.1.4
Humidity - nonoperating	3.3.5	4.6.1.5

4.6.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.7 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.7.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.7.2 Contractor-furnished inspection equipment.

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4.7.2.1 Contractor design. The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

4.7.2.2 Test equipment. In conjunction with 4.5.2.1, the test equipment listed in Table VII shall be used in the performance of the applicable tests as specified in 4.6.

TABLE VII. TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power Supply	20.0 ± .1 Vdc, regulated. AC ripple: Less than 50 mv (millivolt) peak-to-peak. Capacity: 50 ma (milliamp).
2. Power Supply	15.0 ± 1 Vdc, regulated. AC ripple: Less than 50 mv peak-to-peak. Capacity: 50 ma.
3. Power Supply	5.0 ± 0.2 Vdc, regulated. AC ripple: Less than 50 mv peak-to-peak. Capacity: 600 ma.
4. Oscilloscope	Dual channel. Rise time of scope plus measuring probe: 10 ns (nanoseconds) or less. Sweep period: 50 ns/div. or greater. Input cap: Less than 25 pf. Input res: Greater than 8 megohms.
5. Test Fixture	As specified in Figure 2. See Table VIII for test fixture parts list.

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6.	Ohmmeter	To measure 1000 ohms ± 10%.
7.	Digital Voltmeter (DVM)	A range of at least .01 to 100 volts dc. Accuracy of at least ± 0.2 percent of reading. Input impedence of at least 10 megohms on all ranges.
8.	Frequency Counter	Dual input. Time interval B-C function: 1 ms (milliseconds) to 10 sec. Frequency Range: .01 Hz to 100 Hz. Input Sensitivity: Minimum 100 mv RMS each channel, Variable Attenuator 40 dB.

TABLE VIII. Test Fixture Parts List

Designation	Nomenclature and Description
E1, E3, E4, E6	Binding post, H.H. Smith, 257-R or equiv.
E2, E5	Binding post, H.H. Smith, 257-B or equiv.
J1	Connector; Frankford Arsenal, Part No. 10554938
S1, S2, S3, S4	Switch SPDT
TP1-TP20	Tip Jack, H.H. Smith, 1501 or equiv.
R1	Resistor, 4.0K, .1%, 1/2 Watt
R2	Resistor, 5.37K, .1%, 1/2 Watt
R3	Resistor, 1K, 1%, 1/2 Watt

4.8 Test methods and procedures.

4.8.1 Environmental. Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.8.1.1 Vibration. The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810, except that Failure Criteria of 4.9.1.3 does not apply. The frequency and amplitude limits of Paragraph 3.3.1 of this specification shall be used. After vibration, the target simulator shall be tested as given in the Sub-paragraphs of 4.8.2 and 4.8.3 for compliance with 3.3.1.

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4.8.1.2 Shock. This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 g's. After the shock impulses, the Range and Range Rate Generator shall be tested as specified in 4.8.2 and 4.8.3 for compliance with 3.3.2.

4.8.1.3 Altitude (non-operating and operating). This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.8.3.

4.8.1.4 Temperature.

4.8.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The temperature of Step 2 shall be +185°F. The temperature of Step 4 shall be +131°F. In Step 5, the Range and Range Rate Generator shall be tested in accordance with 4.8.2 and 4.8.3.

4.8.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedure 1, Method 502, of MIL-STD-810. The temperature of Step 2 shall be -85°F maintained for 24 hours. The temperature of Step 4 shall be +32°F. In Steps 5 and 7, the Range and Range Rate Generator shall be tested in accordance with 4.8.2 and 4.8.3.

4.8.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure I, Method 507, of MIL-STD-810 except that in Step 4, the relative humidity shall be maintained at 95%.

4.8.2 Electrical.

4.8.2.1 Continuity. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.1.

4.8.2.2 Resistance. Resistance shall be measured using an ohmmeter, Item 6 of Table VII, for compliance to the requirements of 3.4.2.

4.8.3 Performance.

4.8.3.1 Conditions of operation. Unless otherwise specified, all tests in 4.8.3 shall be performed with the D.C. Power Supply connections illustrated in Figure 1. The test fixture, Item 5 of

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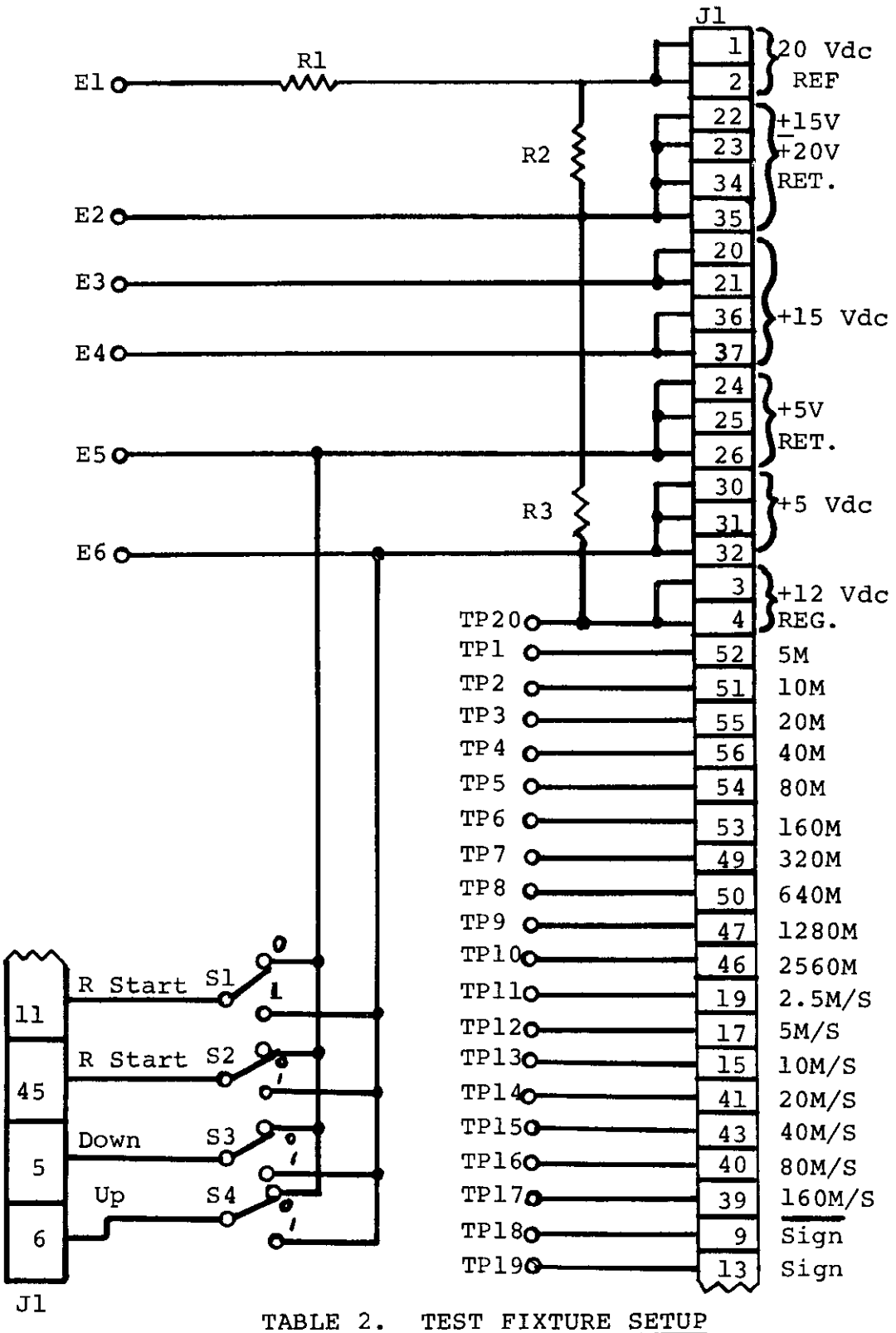


TABLE 2. TEST FIXTURE SETUP

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Table VII, shall be connected to the applicable terminals of the Range and Range Rate Generator, as illustrated in Figure 2. Power supplies 1, 2, and 3 of Table VII shall be connected to the designated terminals of the test fixture in compliance with the test setup of Figure 1. All power supplies shall be adjusted to meet the voltage limits given in Figure 1.

4.8.3.2 Output pin 3. With the conditions of 4.8.3.1 applied, a voltmeter, Item 7 of Table VII, shall be used to measure the voltage between TP20(+) and E2(-) on the test fixture. The voltage shall be checked for compliance to the requirements of 3.5.3.

4.8.3.3 Range output. With the conditions of 4.8.3.1 applied to S1 through S4 set to "0" position, connect Channel 1 of the oscilloscope, Item 4 of Table VII, to TP1 of test fixture, and return to terminal E5 of the test fixture. Determine that a square wave is present at these terminals and that it complies to the amplitude requirements of 3.5.4. In like manner, apply the oscilloscope in turn to TP2 through TP10 with the return to E5 terminal. Determine that the amplitude requirements of 3.5.4 are met at each terminal. In like manner, connect Channel B if the frequency counter, Item 8 of Table VIII, to TP1 through TP10 and verify the period requirements specified for each test point of Table II.

4.8.3.4 Range rate output. With the conditions of 4.8.3.1 applied and S1 through S4 set to "0" position, connect Channel 1 of the oscilloscope, Item 4 of the test fixture to TP11 of the test fixture, and return to terminal E5 of the test fixture. Determine that a square wave is present at these terminals and that it complies to the amplitude requirements of 3.5.5. In like manner, apply the oscilloscope in turn to TP12 through TP17 with the return to E5 terminal. Determine that the amplitude requirements of 3.5.5 are met at each terminal. In like manner, connect Channel B of the frequency counter, Item 8 of Table VII, to TP11 through TP17 and verify the period requirement specified for each test point of Table III.

4.8.3.5 Range count signals.

4.8.3.5.1 Up count range. With the conditions of 4.8.3.1 applied, set S1 through S4 on the test fixture to Step 1 of Table IV. Connect Channel 1 and sync of the oscilloscope, Item 4 of Table VII, to TP1 of the test fixture. Connect Channel 2 of the oscilloscope to TP2 of the test fixture. Connect Channel B of the frequency counter, Item 8 of Table VII, to TP2 of the test fixture and connect Channel C

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of the counter to TP1 of the unit under test. Set the counter function to time B-C and set Channel B trigger to (+) and Channel C trigger to (-). Toggle S1 on the test fixture to "O" and back to "1". Observing the oscilloscope, the signals shall be out of phase till the counter stops counting, then in phase until the run is over. The counter shall verify the time requirement of 3.5.6.1 for the first half of the run. Set the counter Channel "C" trigger to (+) and repeat S1 toggle. The counter shall verify the time requriement of 3.5.6.1 for an entire run.

4.8.3.5.2 Down count range. With the conditions of 4.8.3.1 applied, set S1 through S4 on the test fixture to Step 3 of Table IV. Connect Channel 1 and sync of the oscilloscope to TP1 of the test fixture. Connect Channel 2 of the oscilloscope to TP2 of the test fixture. Connect Channel B of the frequency counter to TP2 of the test fixture and connect Channel C to TP1 on the unit under test. Set the counter function to time B-C and set Channel B trigger to (+) and Channel C trigger to (+). Toggle S1 on the test fixture to "O" and back to "1". Observing the oscilloscope the signals shall be in phase till the counter stops counting, then out of phase until the run is over. The counter shall verify the time requirement of 3.5.6.2 for the first half of the run. Set the counter Channel C trigger to (-) and repeat S1 toggle. The counter shall verify the time requirement of 3.5.6.2 for an entire run.

4.8.3.6 Range rate count signals.

4.8.3.6.1 Up count range rate. With the conditions of 4.8.3.1 applied, Set S1 through S4 on the test fixture to Step 2 of Table IV. Connect Channel 1 and sync of the oscilloscope to TP11 of the test fixture. Connect Channel 2 of the oscilloscope to TP12 of the test fixture. Connect Channel B of the frequency counter to TP12 of the test fixture and connect Channel C to TP1 of the unit under test. Set the counter function to time B-C and set Channel B trigger to (+) and Channel C trigger to (-). Toggle S2 on the test fixture to "O" then back to "1" position. Observing the oscilloscope, the signals must be out of phase till the counter stops counting, then in phase until the run is over. The counter shall verify the time requirement of 3.5.7.1 for the first half of the run. Set the counter Channel C trigger to (+) and repeat first half of the run. Set the counter Channel C trigger to (+) and repeat S1 toggle. The counter shall verify the time requirement of 3.5.7.1 for an entire run.

4.8.3.6.2 Down count range rate. With the conditions of 4.8.3.1 applied, set S1 through S4 on the test fixture to Step 4 of Table IV. Connect Channel 1 and sync of the oscilloscope to TP11 of the test

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fixture. Connect Channel 2 of the oscilloscope to TP12 of the test fixture. Connect Channel B of the frequency counter to TP12 of the test fixture and connect Channel C to TP1 of the unit under test. Set the counter function to time B-C and set Channel B trigger to (+) and Channel C trigger to (+). Toggle S1 on the test fixture to "0" and back to "1" position. Observing the oscilloscope, the signals shall be in phase till the counter stops counting, then out of phase until the run is over. The counter shall verify the time requirement of 3.5.7.2 for the first half of the run. Set the counter Channel C to trigger (-) and repeat S1 toggle. The counter shall verify the time requirement of 3.5.7.2 for an entire run.

4.8.3.7 Sign, $\overline{\text{sign}}$ output. With the conditions of 4.8.3.1 applied, set $\overline{\text{S1 through S4}}$ switches on the test fixture to Step 4 of Table IV. Connect the frequency counter Channel B to TP11 of the test fixture and connect Channel C of the counter to TP1 of the unit under test. Set the counter to time B-C and set Channel B trigger to (+) and Channel C trigger to (+). Connect the digital voltmeter, Item 7 of Table VII, to TP18 of the test fixture and connect the digital voltmeter, Item 7 of Table VII, to TP19 of the test fixture. Toggle S2 on the test fixture to "0" and back to "1" position. The voltmeters shall verify the requirements of 3.5.8 for in phase signals till the counter stops counting, then shall verify the requirements of 3.5.8 for out of phase signals till the end of the run. Set the S1 through S4 switches on the test fixture to Step 2 of Table IV. Set the counter Channel C trigger to (-). Toggle S2 and the voltmeters shall verify the requirements of 3.5.8 for out of phase signals till the counter stops counting, then shall verify the requirements of 3.5.8 for in phase signals until the end of the run.

5. PACKAGING.

5.1 Packaging. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11747101. The level of protection shall be as specified in the contract.

6. NOTES.

6.1 The Range and Range Rate Generator is used to provide range and range rate signals to the SCG Servo Test Panel, to locate defective portions of the mechanical and electronic servo assembly.

6.2 Ordering data. Procurement documents should specify the following:

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a. Title, number, and date of this specification.
b. Applicable packaging data sheet (see 5.1).
c. Selection of applicable level of preservation, packaging, and packing.
d. Level of protection required.
e. Applicable stock number.
f. Block of serial numbers to identify each Range and Range Rate Generator.

6.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring of the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAF-I, Dover, NJ 07801.

When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc.).
c. The designs remaining to be submitted and the expected date of transmittal.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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